

Does Adding Textual Information Improve Remote Sensing Image Classification Performance Compared to Image Only Inputs?

Bahaddin Arda Atar

Informatics Institute, Middle East Technical University (METU), Ankara, Türkiye

Abstract— Multimodal transformers have shown strong performance in vision language tasks, but their effectiveness in structured domains such as remote sensing remains unclear. This study investigates whether textual information improves classification performance compared to image only inputs. A FLAVA-based model is used to compare image only and image+text settings within a unified architecture. Ground truth labels are derived from segmentation masks using pixel-wise class distributions. Proof of concept demonstrates that the pipeline is functional and suitable for further experimentation, although the model is not yet trained.

I. INTRODUCTION

Recent studies have shown that combining images and text, rather than working with only images or only text, can yield more positive results for multimodal learning. Although several models have achieved strong performance in tasks such as image captioning and retrieval, their contribution to classification tasks in structured domains such as remote sensing remains underexplored. Remote sensing datasets frequently comprise both images and textual descriptions. When structured sources like segmentation masks can provide reliable ground truth labels, it is unclear whether textual information provides additional benefit. This leads to the next research question:

“Does incorporating textual information enhance the performance of remote sensing image classification relative to image-only inputs?”

II. LITERATURE SURVEY

Recently, much research has been done on models capable of processing both images and text. CLIP introduced contrastive learning between images and text. This model makes zero-shot classification feasible. [1] BLIP improved vision language understanding by combining caption generation and representation learning through bootstrapped pretraining. It focuses on enhancing alignment via caption refinement. [2]

However, these models primarily focus on alignment rather than unified multimodal processing across different input settings. For this project, a model capable of working with images only, text only, or both images and text is needed. FLAVA is a model that can be used for this purpose. This model introduces a unified transformer architecture that jointly learns from image-only, text-only, and multimodal inputs. [3]

III. MODELING

The dataset includes remote sensing images, corresponding segmentation masks, and captions generated by different

models for these images. Ground truth is derived from the segmentation masks. There is a total of 7 land cover classes, with each pixel representing one of these classes. For the PoC study, the dominant class in the pixel distribution was selected as the target label.

The FLAVA model was chosen for this study due to its ability to process images, text, and both images and text together. Two different configurations were defined: image only and image+text.

IV. FEASIBILITY AND LIMITATIONS

While the literature review supports the study, specific limitations may affect the methodology for this project. Class imbalance in the dataset could impact the model. Furthermore, since texts reflect semantic meaning more, the model might be biased towards text. Also, because the dataset is very large, the transformer model may require significant computational resources. Additionally, reducing segmentation to a single class could lead to information loss.

All these issues will be evaluated during the project's development phase. The PoC study was conducted to demonstrate the feasibility of the pipeline.

V. CONCLUSION

This study proposes a framework to evaluate the effect of textual information, in addition to image information, on image classification. For this purpose, the FLAVA model will be used, and ground truth labels will be obtained from mask images. The Proof-of-Concept (PoC) study demonstrated pipeline feasibility. Subsequent studies will focus on training, benchmarking, and ablation studies.

VI. REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger and I. Sutskever, "Learning Transferable Visual Models From Natural Language Supervision," 2021.
- [2] Li, Junnan; Li, Dongxu; Xiong, Caiming; Hoi, Steven, "for, BLIP: Bootstrapping Language-Image Pre-training Unified Vision-Language Understanding and Generation," 2022.
- [3] A. Singh, R. Hu, V. Goswami, G. Couairon, W. Galuba, M. Rohrbach and D. Kiela, "FLAVA: A Foundational Language And Vision Alignment Model," 2021.